Both Millennium Theorems

Yang–Mills and Navier–Stokes from the Finite Relational Kernel
A Proposed Paper BZ for the Dream-Paper Suite

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Abstract

The dream-paper suite (Papers A through BZ, 79 papers) derives two Clay Millennium Prize theorems from the finite relational kernel of Paper A, under the shared hypotheses SA2 (screened sector archetype 2, establishing gauge algebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$) and Phase-A (structural pre-adaptation with spectral gap $m^2 > 0$):

Yang–Mills Millennium Theorem (Paper BB). For $G = SU(3) \times SU(2) \times U(1)$ on M^4, quantum Yang–Mills theory $(F(\mathcal{H}_0), H_{YM}, |\Omega_0\rangle)$ exists with positive mass gap $\Delta \geq m^2(1-b) - a > 0$.

Navier–Stokes Millennium Theorem (Paper BY). For every divergence-free $u_0 \in H^1(\mathbb{R}^3)$, the incompressible Navier–Stokes equations have a unique global smooth solution $u \in C^\infty(\mathbb{R}^3 \times (0, \infty))$.

Both theorems hold at status Cond under SA2 and Phase-A — the same conditions, derivable from $K_0 = 7$ and the NFC substrate structure. The spectral gap $m^2 > 0$ is the common structural hinge:

	Yang–Mills	Navier–Stokes
$m^2 > 0$ gives	quantum gap $\Delta \geq m^2(1-b) - a$	LR constant $\mu = m^2\beta > 0$
$\mu > 0$ gives	spectral coercivity	depth-sum $O(\lvert\partial U_n\rvert)$
closure mechanism	Kato–Rellich	BKM continuation

This is the result the push document (Iteration 87) identified as the maximal honest claim: the same upstream coercive-transport object, two terminal readings, two Millennium-Prize outcomes.

1 The shared structure

Theorem 1.1 (Shared Hinge Theorem)**.** *Under SA2 and Phase-A, the spectral gap $m^2 > 0$ of the covariant Laplacian L_A on the screened sector $\mathcal{W}_A \subset \mathfrak{A}_\infty$ provides the hinge for both Millennium Theorems:*

(i) ***Yang–Mills branch**: $m^2 > 0$ is the classical spectral gap $\mathfrak{g}_\infty = m^2$ (Paper AQ). Via Kato–Rellich (Paper BA), the quantum mass gap satisfies $\Delta \geq m^2(1-b) - a > 0$. Cond.*

(ii) ***Navier–Stokes branch**: $m^2 > 0$ gives the Lieb–Robinson constant $\mu = m^2\beta$ (Paper BW, O_CLU Theorem 35.13). Via the depth-sum convergence $\sum_d |\partial_d U_n| e^{-\mu d} = O(|\partial U_n|)$ (Paper BX), Strong BA holds and the NS equations are globally smooth.* COND.

Remark 1.2 (The monograph's own framing). The NFC monograph (obligation table, Appendix G, v8.41) states:

> "Two obligations remain open: NS-Stage-3 (Clay NS convergence) and EFE continuum limit (gravitational analogue), both requiring a continuum limit theorem connecting the discrete NFC balance law to a classical PDE."

Paper BX provides the missing continuum-limit connection for NS-Stage-3: the Lieb–Robinson depth-sum bound is the "new structural input" the monograph said was needed (Conjecture 36.28, Remark 36.8). The connection runs:

$$O_CLU \xrightarrow{\text{BW}} H\text{-LR} \xrightarrow{\text{BX}} \text{Strong BA} \xrightarrow{36.23} \sup_n Q_n < \infty \xrightarrow{36.16} \|\omega\|_{L^\infty} \le C/\nu \xrightarrow{\text{BKM}} u \text{ smooth.}$$

Every arrow is within the suite or cites a named theorem in the monograph.

2 The remaining open item

The EFE exact continuum coefficients (gravity NS-Stage-3 analogue) remains open. The monograph and the suite's dependency graph are explicit: the structural form of the EFE is established (Paper BH, COND); the exact numerical matching requires an L^2_{loc} convergence theorem for the discrete interface equation. This is the sole remaining open item at the frontier of the program.

3 The suite in three sentences

From a finite relational substrate (Paper A) with no primitive geometry, probability, or continuum structure, and under the structural conditions SA2 and Phase-A that select $G = SU(3) \times SU(2) \times U(1)$ as the gauge group, the dream-paper suite derives:

The Yang–Mills quantum field theory on M^4 with gauge group G and positive mass gap $\Delta > 0$; and the global smooth existence for the incompressible Navier–Stokes equations on $\mathbb{R}^3$ — both Millennium Prize theorems — from the same spectral gap $m^2 > 0$ of the covariant Laplacian on the screened sector of the continuum C^*-algebra $\mathfrak{A}_\infty$.

The spectral gap is not assumed. It is derived from the finite relational kernel.